



Hubble images cartwheel

Hubble has captured an image of the Cartwheel galaxy, which are actually two colliding galaxies.
<http://digit.in/glxwheel>



Hypatia stone puzzles scientists

A mysterious meteorite embedded with diamonds was possibly formed before the solar system.
<http://digit.in/hypatiastone>

An artist's impression of 'Oumuamua.

Credit: ESO/M. Kornmesser

'OUMUAMUA: THE INTERSTELLAR ROLLING STONE

An asteroid ejected from the an alien system zipped past the Sun, making astronomers around the world rush to study the object before it travelled too far to observe.

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In October, astronomers working with the Pan-STARRS1 telescope came across a highly unusual object in the Solar System. The object had been wandering between the stars for millions of years before encountering the Solar System, entirely by chance. While scientists had hypothesised the existence of such objects, they had never directly observed one.

The International Astronomical Union had to rush to come up with a new naming convention for the unique object. The object was named 'Oumuamua, based on the recommendations of the team that made the discovery. For the pronunciation, just add an "h" sound before the O. The name means a messenger from

the distant past. Apart from its extra-solar origins, the shape of the object was extremely elongated.

The discovery and the shape of 'Oumuamua, was very similar to a mysterious interstellar object in the Arthur C Clarke science fiction novel, *Rendezvous with Rama*. Olivier Hainaut from the European Southern Observatory even said in a press release, "This felt very much like the beginning of the famous science fiction story, *Rendezvous with Rama*." Astronomer Jason Wright from Penn State noted on his blog, AstroWright, "Its discovery closely tracks the opening chapter of the book *Rendezvous with Rama*, by Arthur C. Clarke, about the discovery of an interstellar spaceship on a similar trajectory to 'Oumuamua."

One paragraph that equally applies to 'Oumuamua and Rama is "It was a lonely wanderer between the stars, making its first and last visit to the Solar System – for it was moving so swiftly that the gravitational field

of the sun could never capture it. It would flash inwards past the orbits of Jupiter, Mars, Earth, Venus and Mercury, gaining speed as it did so, until it rounded the sun and headed out once again into the unknown."

The asteroid appears to have come from the direction of the star Vega in the constellation of Lyra. However, when the asteroid was in that location 3,00,000 years ago, the constellation appeared in a different place from Earth. At the time of the discovery, 'Oumuamua had already crossed the point closest to the Sun, and was heading towards the constellation of Pegasus.

Unlike Clarke's book, astronomers actually interrupted the planned observation campaigns of telescopes to take a closer look at the curious interstellar nomad. There was very limited time on hand, it was highly unlikely that the object would ever visit the Solar System again, and researchers could only guess when an opportunity for studying an interstellar asteroid would arise again. For all these reasons, astronomers jumped at the chance to observe the asteroid. A group of astronomers used three telescopes in Hawaii and two telescopes in Chile to conduct follow on observations of 'Oumuamua, in an effort to study its properties, and reveal its mysteries. Another group of astronomers studied the object through telescopes in Hawaii and Spain. A third team of international astronomers studied the object from a telescope in New Mexico. NASA's Hubble and Spitzer space telescopes also tracked the asteroid. Yan Fernandez from the University of Central Florida said, "On very, very short notice they were able to rework the telescope's schedule — which is set months in advance — to get us 4 hours of time". Director of the Gemini Observatory, Laura Ferrarese said, "Needless to say, we dropped everything so we could quickly point the Gemini telescopes at this object immediately after its discovery."

The findings were published in a paper titled "A brief visit from a red